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System, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system

Abstract

The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. In one embodiment, the system receives search-related signals from a plurality of signal sources. The signals are indicative of: (1) a user's intent to perform a search for event logs in a cybersecurity database, (2) how computationally intensive the potential search is likely to be, and (3) the currently available computational resources. The signals are evaluated, and an autoscale prediction score is calculated. The autoscale prediction score reflects the likelihood of a user to submit search, the computational resources required for the potential search, and the currently available computational resources. The system scales computational resources in accordance with the autoscale prediction score. These steps are performed before any search is submitted by the user in a search user interface.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5941947	12/1998	Brown et al.	N/A	N/A
6223985	12/2000	DeLude	N/A	N/A
6594481	12/2002	Johnson et al.	N/A	N/A
7181768	12/2006	Ghosh et al.	N/A	N/A
7624277	12/2008	Simard et al.	N/A	N/A
7668776	12/2009	Ahles	N/A	N/A
8326788	12/2011	Allen et al.	N/A	N/A
8443443	12/2012	Nordstrom et al.	N/A	N/A
8479302	12/2012	Lin	N/A	N/A
8484230	12/2012	Harnett et al.	N/A	N/A
8539088	12/2012	Zheng	N/A	N/A
8583781	12/2012	Raleigh	N/A	N/A
8606913	12/2012	Lin	N/A	N/A
8676273	12/2013	Fujisake	N/A	N/A
8850570	12/2013	Ramzan	N/A	N/A
8881289	12/2013	Basavapatna et al.	N/A	N/A
9055093	12/2014	Borders	N/A	N/A
9081958	12/2014	Ramzan et al.	N/A	N/A
9129110	12/2014	Mason et al.	N/A	N/A
9185095	12/2014	Moritz et al.	N/A	N/A
9189623	12/2014	Lin et al.	N/A	N/A
9202052	12/2014	Fang et al.	N/A	N/A
9680938	12/2016	Gil et al.	N/A	N/A
9690938	12/2016	Saxe et al.	N/A	N/A
9692765	12/2016	Choi et al.	N/A	N/A
9760240	12/2016	Maheshwari et al.	N/A	N/A
9779253	12/2016	Mahaffey et al.	N/A	N/A
9798883	12/2016	Gil et al.	N/A	N/A
9832138	12/2016	Skalski	N/A	G06F 9/5083
9843596	12/2016	Averbuch et al.	N/A	N/A
9898604	12/2017	Fang et al.	N/A	N/A
10063582	12/2017	Feng et al.	N/A	N/A
10095871	12/2017	Gil et al.	N/A	N/A
10178108	12/2018	Lin et al.	N/A	N/A
10354015	12/2018	Kalchbrenner et al.	N/A	N/A
10360387	12/2018	Jou et al.	N/A	N/A
10397272	12/2018	Bruss et al.	N/A	N/A
10419470	12/2018	Segev et al.	N/A	N/A

10445311	12/2018	Saurabh et al.	N/A	N/A
10467631	12/2018	Dhurandhar et al.	N/A	N/A
10474828	12/2018	Gil et al.	N/A	N/A
10496815	12/2018	Steiman et al.	N/A	N/A
10621343	12/2019	Maciejak et al.	N/A	N/A
10645109	12/2019	Lin et al.	N/A	N/A
10685293	12/2019	Heimann et al.	N/A	N/A
10803183	12/2019	Gil et al.	N/A	N/A
10819724	12/2019	Amiri et al.	N/A	N/A
10841338	12/2019	Lin et al.	N/A	N/A
10887325	12/2020	Lin et al.	N/A	N/A
10944777	12/2020	Lin et al.	N/A	N/A
11017173	12/2020	Lu et al.	N/A	N/A
11080483	12/2020	Islam et al.	N/A	N/A
11080591	12/2020	van den Oord et al.	N/A	N/A
11128600	12/2020	Bansal et al.	N/A	N/A
11140167	12/2020	Lin et al.	N/A	N/A
11151471	12/2020	Niininen et al.	N/A	N/A
11178168	12/2020	Lin et al.	N/A	N/A
11245716	12/2021	Roelofs et al.	N/A	N/A
11423143	12/2021	Lin et al.	N/A	N/A
11431741	12/2021	Lin et al.	N/A	N/A
11463331	12/2021	Arlitt et al.	N/A	N/A
11625366	12/2022	Steiman et al.	N/A	N/A
11736527	12/2022	Durairaj et al.	N/A	N/A
11843505	12/2022	Cruise	N/A	G06F 16/953
11956253	12/2023	Lin et al.	N/A	N/A
12034732	12/2023	Lin et al.	N/A	N/A
12063226	12/2023	Lin et al.	N/A	N/A
12164402	12/2023	Xie	N/A	G06F 11/3006
2002/0107926	12/2001	Lee	N/A	N/A
2003/0065926	12/2002	Schultz et al.	N/A	N/A
2003/0147512	12/2002	Abhuri	N/A	N/A
2004/0073569	12/2003	Knott et al.	N/A	N/A
2006/0090198	12/2005	Aaron	N/A	N/A
2007/0156771	12/2006	Hurley et al.	N/A	N/A
2007/0282778	12/2006	Chan et al.	N/A	N/A
2008/0028467	12/2007	Kommareddy et al.	N/A	N/A
2008/0040802	12/2007	Pierson et al.	N/A	N/A
2008/0170690	12/2007	Tysowski	N/A	N/A
2008/0262990	12/2007	Kapoor et al.	N/A	N/A
2008/0301780	12/2007	Ellison et al.	N/A	N/A
2009/0144095	12/2008	Shahi et al.	N/A	N/A
2009/0171752	12/2008	Galvin et al.	N/A	N/A
2009/0292954	12/2008	Jiang et al.	N/A	N/A
2009/0293121	12/2008	Bigus et al.	N/A	N/A
2010/0125911	12/2009	Bhaskaran	N/A	N/A
2010/0191763	12/2009	Wu	N/A	N/A
2010/0269175	12/2009	Stolfo et al.	N/A	N/A
2010/0284282	12/2009	Golic	N/A	N/A

2011/0167495	12/2010	Antonakakis et al.	N/A	N/A
2012/0278021	12/2011	Lin et al.	N/A	N/A
2012/0316835	12/2011	Maeda et al.	N/A	N/A
2012/0316981	12/2011	Hoover et al.	N/A	N/A
2013/0080631	12/2012	Lin	N/A	N/A
2013/0086273	12/2012	Wray	709/226	G06F 9/5072
2013/0117554	12/2012	Ylonen	N/A	N/A
2013/0197998	12/2012	Buhrmann et al.	N/A	N/A
2013/0227643	12/2012	Mccoog et al.	N/A	N/A
2013/0268260	12/2012	Lundberg et al.	N/A	N/A
2013/0305357	12/2012	Ayyagari et al.	N/A	N/A
2013/0340028	12/2012	Rajagopal et al.	N/A	N/A
2014/0007238	12/2013	Magee	N/A	N/A
2014/0090058	12/2013	Ward et al.	N/A	N/A
2014/0101759	12/2013	Antonakakis et al.	N/A	N/A
2014/0315519	12/2013	Nielsen	N/A	N/A
2014/0365418	12/2013	Grant	N/A	N/A
2015/0026027	12/2014	Priess et al.	N/A	N/A
2015/0039543	12/2014	Athmanathan et al.	N/A	N/A
2015/0046969	12/2014	Abuelsaad et al.	N/A	N/A
2015/0100558	12/2014	Fan	N/A	N/A
2015/0121503	12/2014	Xiong	N/A	N/A
2015/0205944	12/2014	Turgeman	N/A	N/A
2015/0215325	12/2014	Ogawa	N/A	N/A
2015/0339477	12/2014	Abrams et al.	N/A	N/A
2015/0341379	12/2014	Lefebvre et al.	N/A	N/A
2015/0363691	12/2014	Gocek et al.	N/A	N/A
2016/0005044	12/2015	Moss et al.	N/A	N/A
2016/0021117	12/2015	Harmon et al.	N/A	N/A
2016/0063397	12/2015	Ylipaavalniemi et al.	N/A	N/A
2016/0292592	12/2015	Patthak et al.	N/A	N/A
2016/0306965	12/2015	Iyer et al.	N/A	N/A
2016/0364427	12/2015	Wedgeworth, III	N/A	N/A
2017/0019506	12/2016	Lee et al.	N/A	N/A
2017/0024135	12/2016	Christodorescu et al.	N/A	N/A
2017/0127016	12/2016	Yu et al.	N/A	N/A
2017/0155652	12/2016	Most et al.	N/A	N/A
2017/0161451	12/2016	Weinstein et al.	N/A	N/A
2017/0178026	12/2016	Thomas et al.	N/A	N/A
2017/0213025	12/2016	Srivastav et al.	N/A	N/A
2017/0223035	12/2016	Watanabe	N/A	H04L 41/0896
2017/0236081	12/2016	Grady Smith et al.	N/A	N/A
2017/0264679	12/2016	Chen et al.	N/A	N/A
2017/0318034	12/2016	Holland et al.	N/A	N/A
2017/0322959	12/2016	Tidwell	N/A	G06F 21/554
2017/0323636	12/2016	Xiao et al.	N/A	N/A
2018/0004961	12/2017	Gil et al.	N/A	N/A
2018/0039699	12/2017	Wan	N/A	G06F 16/3322
2018/0048530	12/2017	Nikitaki et al.	N/A	N/A
2018/0063168	12/2017	Sofka	N/A	N/A

2018/0069893	12/2017	Amit et al.	N/A	N/A
2018/0075343	12/2017	van den Oord et al.	N/A	N/A
2018/0089304	12/2017	Vizer et al.	N/A	N/A
2018/0097822	12/2017	Huang et al.	N/A	N/A
2018/0144139	12/2017	Cheng et al.	N/A	N/A
2018/0157963	12/2017	Salti et al.	N/A	N/A
2018/0165554	12/2017	Zhang et al.	N/A	N/A
2018/0181883	12/2017	Ikeda	N/A	N/A
2018/0190280	12/2017	Cui et al.	N/A	N/A
2018/0234443	12/2017	Wolkov et al.	N/A	N/A
2018/0248895	12/2017	Watson et al.	N/A	N/A
2018/0285340	12/2017	Murphy et al.	N/A	N/A
2018/0288063	12/2017	Koottayi et al.	N/A	N/A
2018/0288086	12/2017	Amiri et al.	N/A	N/A
2018/0307994	12/2017	Cheng et al.	N/A	N/A
2018/0316701	12/2017	Holzhauer et al.	N/A	N/A
2018/0322368	12/2017	Zhang et al.	N/A	N/A
2019/0014149	12/2018	Cleveland et al.	N/A	N/A
2019/0028496	12/2018	Fenoglio et al.	N/A	N/A
2019/0034641	12/2018	Gil et al.	N/A	N/A
2019/0066185	12/2018	More et al.	N/A	N/A
2019/0080225	12/2018	Agarwal	N/A	N/A
2019/0081957	12/2018	Thrash et al.	N/A	N/A
2019/0089721	12/2018	Pereira et al.	N/A	N/A
2019/0103091	12/2018	Chen	N/A	N/A
2019/0114419	12/2018	Chistyakov et al.	N/A	N/A
2019/0122078	12/2018	Ura	N/A	G06N 20/20
2019/0124045	12/2018	Zong et al.	N/A	N/A
2019/0124093	12/2018	Sharma et al.	N/A	N/A
2019/0132629	12/2018	Kendrick	N/A	N/A
2019/0149565	12/2018	Hagi et al.	N/A	N/A
2019/0171655	12/2018	Psota et al.	N/A	N/A
2019/0173804	12/2018	Nicas	N/A	H04L 47/822
2019/0182280	12/2018	La Marca et al.	N/A	N/A
2019/0205750	12/2018	Zheng et al.	N/A	N/A
2019/0207969	12/2018	Brown	N/A	N/A
2019/0213247	12/2018	Pala et al.	N/A	N/A
2019/0244603	12/2018	Angkititrakul et al.	N/A	N/A
2019/0303703	12/2018	Kumar et al.	N/A	N/A
2019/0318100	12/2018	Bhatia et al.	N/A	N/A
2019/0334784	12/2018	Kvernvik et al.	N/A	N/A
2019/0349400	12/2018	Bruss et al.	N/A	N/A
2019/0378051	12/2018	Widmann et al.	N/A	N/A
2019/0384762	12/2018	Hill	N/A	G06F 16/2272
2020/0021607	12/2019	Muddu et al.	N/A	N/A
2020/0021620	12/2019	Purathepparambil et al.	N/A	N/A
2020/0034481	12/2019	Asplund	N/A	G06F 16/35
2020/0082098	12/2019	Gil et al.	N/A	N/A
2020/0137104	12/2019	Hassanzadeh et al.	N/A	N/A
2020/0177618	12/2019	Hassanzadeh et al.	N/A	N/A

2020/0228557	12/2019	Lin et al.	N/A	N/A
2020/0302118	12/2019	Cheng et al.	N/A	N/A
2020/0327886	12/2019	Shalaby et al.	N/A	N/A
2021/0081459	12/2020	Chung	N/A	H04L 51/224
2021/0089884	12/2020	Macready et al.	N/A	N/A
2021/0125050	12/2020	Wang	N/A	N/A
2021/0126938	12/2020	Trost et al.	N/A	N/A
2021/0133331	12/2020	Lipkis et al.	N/A	N/A
2021/0182612	12/2020	Zeng et al.	N/A	N/A
2021/0232768	12/2020	Ling et al.	N/A	N/A
2021/0248240	12/2020	Comish et al.	N/A	N/A
2022/0006814	12/2021	Lin et al.	N/A	N/A
2022/0030017	12/2021	Infante-Lopez et al.	N/A	N/A
2022/0076164	12/2021	Conort	N/A	G06N 20/00
2022/0147622	12/2021	Chesla	N/A	N/A
2022/0245093	12/2021	Batsakis	N/A	G06F 16/144

OTHER PUBLICATIONS

Bahnsen, Alejandro Correa “Classifying Phishing URLs Using Recurrent Neural Networks”, IEEE 2017, 8 pages. cited by applicant

Chen, Jinghui, et al., “Outlier Detection with Autoencoder Ensembles”, Proceedings of the 2017 SIAM International Conference on Data Mining, pp. 90-98. cited by applicant

Cooley, R., et al., “Web Mining: Information and Pattern Discovery on the World Wide Web”, Proceedings Ninth IEEE International Conference on Tools with Artificial Intelligence, Nov. 3-8, 1997, pp. 558-567. cited by applicant

DatumBox Blog, “Machine Learning Tutorial: The Naïve Bayes Text Classifier”, DatumBox Machine Learning Blog and Software Development News, Jan. 2014, pp. 1-11. cited by applicant

Fargo, Farah “Resilient Cloud Computing and Services”, PhD Thesis, Department of Electrical and Computer Engineering, University of Arizona, 2015, pp. 1-115. cited by applicant

Freeman, David, et al., “Who are you? A Statistical Approach to Measuring User Authenticity”, NDSS, Feb. 2016, pp. 1-15. cited by applicant

Goh, Jonathan et al., “Anomaly Detection in Cyber Physical Systems using Recurrent Neural Networks”, IEEE 2017, pp. 140-145. cited by applicant

Guo, Diansheng et al., “Detecting Non-personal and Spam Users on Geo-tagged Twitter Network”, Transactions in GIS, 2014, pp. 370-384. cited by applicant

Ioannidis, Yannis, “The History of Histograms (abridged)”, Proceedings of the 29th VLDB Conference (2003), pp. 1-12. cited by applicant

Kim, Jihyun et al., “Long Short Term Memory Recurrent Neural Network Classifier for Intrusion Detection”, IEEE 2016, 5 pages. cited by applicant

Malik, Hassan, et al., “Automatic Training Data Cleaning for Text Classification”, 11th IEEE International Conference on Data Mining Workshops, 2011, pp. 442-449. cited by applicant

Mietten, Markus et al., “ConXsense-Automated Context Classification for Context-Aware Access Control”, Asia CCS'14, 2014, pp. 293-304. cited by applicant

Poh, Norman, et al., “EER of Fixed and Trainable Fusion Classifiers: A Theoretical Study with Application to Biometric Authentication Tasks”, Multiple Classifier Systems, MCS 2005, Lecture Notes in Computer Science, vol. 3541, pp. 1-11. cited by applicant

Shi, Yue et al., “Cloudlet Mesh for Securing Mobile Clouds from Intrusions and Network Attacks”, 2015 3rd IEEE International Conference on Mobile Cloud Computing, Services, and Engineering, pp. 109-118. cited by applicant

Taylor, Adrian et al., “Anomaly Detection in Automobile Control Network Data with Long Short-

Term Memory Networks”, IEEE 2016, pp. 130-139. cited by applicant
Taylor, Adrian “Anomaly-Based Detection of Malicious Activity in In-Vehicle Networks”, Ph.D. Thesis, University of Ottawa 2017, 151 pages. cited by applicant
Wang, Alex Hai, “Don't Follow Me Spam Detection in Twitter”, International Conference on Security and Cryptography, 2010, pp. 1-10. cited by applicant
Wang, Shuhao et al., “Session-Based Fraud Detection in Online E-Commerce Transactions Using Recurrent Neural Networks”, 2017, 16 pages. cited by applicant
Zhang, Ke et al., “Automated IT System Failure Prediction: A Deep Learning Approach”, IEEE 2016, pp. 1291-1300. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates generally to cybersecurity systems, and, more specifically, to predictive autoscaling for query searches on a database of event logs in a cybersecurity system.

2. Description of the Background Art

(2) Cybersecurity systems monitor entity behavior in a network in order to detect cybersecurity threats. As entities interact with the network, various systems generate raw logs related to the entity behavior. For example, a cybersecurity system may obtain raw data logs related to a user's interactions with the IT infrastructure, such as user logon events, server access events, application access events, and data access events. Cybersecurity system will typically take these raw data logs and generate event logs from the raw data logs.

(3) Cybersecurity analysts within an enterprise perform searches on the event logs to provide a better understanding of cybersecurity risks. Event log databases can be large, and search queries on the event log databases are often compute intensive processes. A query life cycle typically has the following stages: Query Authoring Query Parsing Autoscaling of Compute Resources (horizontal scaling) Query Execution Query Results Downscaling of Compute Resources

(4) Many cybersecurity systems store event logs in cloud-based databases, and scale compute resources for searches on demand. The scaling is performed after the query is submitted and parsed. Dynamically scaling compute resources on demand is more cost efficient than maintaining a high amount of compute resources at all times. Unfortunately, the time required to autoscale compute resources creates a delay in query execution, which results in slow queries. Therefore, there is demand for a solution that can autoscale without slowing down searches.

SUMMARY OF THE DISCLOSURE

(5) The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. The system receives one or more signals indicative of a user's intent to perform a search for event logs in a cybersecurity database. Examples of the signals include a user clicking on a query builder window within a search interface for the event logs database or selecting a previous search. The system calculates an autoscale prediction score for a potential search based on the one or more signals. The system then makes an autoscale decision based on the autoscale prediction score. These steps are all performed prior to a user submitting a search query. In other words, the system makes an autoscale decision based on the likelihood that a user will submit a search. This way, if and when the user does submit

a search, the computational resources for the search are available and there is no delay associated with provisioning additional compute resources.

(6) In a preferred embodiment, the system receives signals from a plurality of different sources that collectively indicate not only the user's intent to perform a search, but also indicate how computationally intensive the search is likely to be and the currently available computational resources. For example, a signal indicating that a user has selected a query time domain within the search user interface provides an indication of the size of the search domain and hence the amount of computational resources required.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a flowchart that illustrate method for predictive autoscaling for event log searches in accordance with one embodiment.
- (2) FIG. 2 is a flowchart that illustrates an example implementation of the method of FIG. 1.
- (3) FIG. 3 is a block diagram that illustrates an example architecture for a system for predictive autoscaling.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (4) The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. The methods disclosed herein are performed by a computer system (“the system”), such as cybersecurity system that detects cybersecurity threats in a network.
- (5) FIG. 1 illustrates a method for predictive autoscaling for event log searches in accordance with one embodiment. The system receives one or more signals indicative of a user's intent to perform a search for event logs in a cybersecurity database (step 110). Examples of the signals include a user clicking on a query builder window within a search interface for the event logs database or selecting a previous search. The system calculates an autoscale prediction score for a potential search based on the one or more signals (step 120). The system then makes an autoscale decision based on the autoscale prediction score (step 130). These steps are all performed prior to a user submitting a search query. In other words, the system makes an autoscale decision based on the likelihood that a user will submit an event log search. This way, if and when the user does submit a search, the computational resources for the search are available, and there is no delay associated with provisioning additional compute resources.
- (6) FIG. 2 illustrates an example implementation of the method of FIG. 1. The system receives signals from a plurality of sources that are collectively indicative of: (1) user intent to perform a search, (2) likely search domain size, and (3) currently available computational resources (step 210). The signals related to the user intent and to the search domain size include signals about a user's actions on a search user interface before the user submits a search. Examples of the signals are set forth below in Table 1.
- (7) The system evaluates the signals and assigns a value to each signal (steps 220-230). The table below lists examples of the signals received and the values which may be assigned to each signal:
- (8) TABLE-US-00001
- | TABLE 1 | What the signal is | Example | Range of Signal | indicative of | Values |
|---------|----------------------|---|----------------------------------|---|--------|
| 1 | User is using a | User intent to search | 0 (no usage), query builder on a | 1 (usage) search user interface (e.g., user clicks on a query bar). | |
| 2 | User selects a query | User intent to search; 0 (no selection of a time window (e.g., Size of search window), time range selection) domain (i.e., 0.1 (selection of a computational “small” window), resources required 0.2 (selection of a for search) “medium” window), and 0.3 (selection of a “large” window). Small, medium, and large are defined by time window ranges. | | | |
| 3 | User selection of a | User intent 0 (no selection), recent query 1 (selection) | | | |
| 4 | Pattern of usage | User intent 0 (current time is hours for the given outside the user's user peak | | | |

search timeframe), 1 (current time is within the user's peak search timeframe) 5 Pattern of usage User intent 0 (current time is hours for the given outside the entity's customer entity peak usage window for searches), 1 (current time is within the entity's peak usage window for searches) 6 Adding indexed User intent to search 0 (an index field is fields to the query and computational not yet included builder resources required within the search criteria), 1 (an indexed field has been added to the search criteria) 7 Percentage of Available compute 0.3 (no compute compute resources resources engines currently available available), 0.2 (small amount of compute engines available), 0.1 (medium amount of compute engines available), 0 (large amount of compute engines available). Small, medium, and large values may be predefined in configuring the system. 8 Size of customer Size of search 0.1 (small), event log database domain 0.2 (medium), 0.3 (large). Small, medium, and large values are defined by data size ranges.

(9) The system calculates an autoscale prediction score as a weighted sum of the values assigned to the signals (step **240**). The score reflects the probability of a user submitting a search, the size of the search domain, and currently available computational resources. The autoscale prediction score may be expressed mathematically as follows:

Autoscale prediction score(APS)= $s_1*w_1+s_2*w_2+s_3*w_3+s_4*w_4+ \dots s_N*w_N$, Where N is the number of signals.

(10) In one embodiment, the signal values and the weights are assigned such that the autoscale prediction score is in the range between 0 and 1. The example signal values in Table 1 are designed for a prediction score between 0 and 1.

(11) The system adjusts computational resources available for searches as a function of the autoscale prediction score (step **250**). The autoscale prediction score is used to decide whether to autoscale and the capacity desired. This may be expressed mathematically as followed: If $APS > \text{threshold value}$, Autoscaler initiates autoscaling. Capacity required= $f(APS)$ $f(APS)$ translates the autoscale prediction score to the amount of compute resources to be provisioned by the Autoscaler.

(12) Steps **210-250** are performed before a user submits a search.

(13) FIG. 3 illustrates an example architecture of the system. A Prediction Service module **320** receives signals from a plurality of signal sources **310**. The Prediction Service Module **320** calculates an autoscale prediction score in accordance with the methods described herein. The Autoscaler **330** provisions compute engines **340** for searching as a function of the autoscale prediction score.

(14) The methods described with respect to FIGS. 1-3 are embodied in software and performed by a computer system (comprising one or more computing devices) executing the software. A person skilled in the art would understand that a computer system has one or more memory units, disks, or other physical, computer-readable storage media for storing software instructions, as well as one or more processors for executing the software instructions.

(15) As will be understood by those familiar with the art, the invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Accordingly, the above disclosure is intended to be illustrative, but not limiting, of the scope of the invention.

Claims

1. A non-transitory computer-readable medium comprising a computer program, that, when executed by a computer system, enables the computer system to perform the following method for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the method comprising: receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system; calculating an autoscale prediction score for a potential search based on the signal; and making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential

search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.

2. The non-transitory computer-readable medium of claim 1, wherein: a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score; the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and the autoscale prediction score is calculated based on the plurality of signals.
3. The non-transitory computer-readable medium of claim 2, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.
4. The non-transitory computer-readable medium of claim 3, wherein the signal indicative of the computational resources required for the potential search is one of the following: selection of a query time domain, size of event log database upon which the potential search would be performed, and addition of an indexed field to a query builder.
5. The non-transitory computer-readable medium of claim 2, wherein the signal indicative of the user's intent to perform the search is one of the following: usage of a query builder for the event log search, selection of a recent query, the user's past patterns of searches, selection of a query time domain, and an addition of an indexed field to a query builder.
6. The non-transitory computer-readable medium of claim 2, wherein calculating the autoscale prediction score comprises: assigning a value to each of the plurality of signals; and calculating a weighted sum of the signal values.
7. The non-transitory computer-readable medium of claim 1, further comprising: receiving a new signal related to the potential search after the autoscale prediction score has been calculated for the potential search; recalculating the autoscale prediction score to factor in the new signal; and making a new autoscale decision based on the recalculated autoscale prediction score.
8. The non-transitory computer-readable medium of claim 1, wherein making an autoscale decision comprises adjusting computational resources available for the potential search.
9. The non-transitory computer-readable medium of claim 8, wherein computational resources are increased as a function of the autoscale prediction score.
10. A computer system for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the system comprising: one or more processors; one or more memory units coupled to the one or more processors, wherein the one or more memory units store instructions that, when executed by the one or more processors, cause the system to perform the operations of: receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system; calculating an autoscale prediction score for a potential search based on the signal; and making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.
11. The system of claim 10, wherein: a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score; the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and the autoscale prediction score is calculated based on the plurality of signals.
12. The system of claim 11, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.
13. The system of claim 12, wherein the signal indicative of the computational resources required for the potential search is one of the following: selection of a query time domain, size of event log database upon which the potential search would be performed, and addition of an indexed field to a

query builder.

14. The system of claim 11, wherein the signal indicative of the user's intent to perform the search is one of the following: usage of a query builder for the event log search, selection of a recent query, the user's past patterns of searches, selection of a query time domain, and an addition of an indexed field to a query builder.

15. The system of claim 11, wherein calculating the autoscale prediction score comprises: assigning a value to each of the plurality of signals; and calculating a weighted sum of the signal values.

16. The system of claim 10, further comprising: receiving a new signal related to the potential search after the autoscale prediction score has been calculated for the potential search; recalculating the autoscale prediction score to factor in the new signal; and making a new autoscale decision based on the recalculated autoscale prediction score.

17. The system of claim 10, wherein making an autoscale decision comprises adjusting computational resources available for the potential search.

18. The system of claim 17, wherein computational resources are increased as a function of the autoscale prediction score.

19. A method, performed by a computer system, for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the method comprising: receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system; calculating an autoscale prediction score for a potential search based on the signal; and making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.

20. The method of claim 19, wherein: a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score; the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and the autoscale prediction score is calculated based on the plurality of signals.

21. The method of claim 20, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.
